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Multi-Algorithm Puzzle Solver: Project Update Report

1. Progress Since Previous Update

After the previous project update, the team focused on completing the remaining implementation tasks and improving the overall system integration. The project progressed from an early prototype into a more complete Artificial Intelligence based puzzle-solving framework.

The major improvements during this period include completing puzzle integration, expanding search algorithm support, developing visualization features, and introducing a performance evaluation framework.

2. Major Development Improvements

The system was extended to support multiple puzzle environments through a common framework.

• 8-Puzzle Solver:
Improved state transition handling and goal verification for generating valid solutions.

• Maze Solver:
Completed grid-based representation and path generation for efficient route searching.

• N-Queens Solver:
Implemented constraint-based modelling to generate valid queen placements.

The search module was enhanced by integrating BFS, DFS, UCS, Greedy Best First Search, and A*. Additional improvements include solution path reconstruction, improved algorithm execution control, and performance tracking.

3. Visualization and Performance Analysis

A major improvement was the development of the visualization framework. Previously, the system mainly focused on generating solutions; the updated version now provides graphical interpretation of algorithm behaviour.

Implemented visualization:
• 8-Puzzle: Displays tile configuration and movement sequence.
• Maze Solver: Shows obstacles, route, start, and goal positions.
• N-Queens: Displays the final chessboard arrangement.

A performance evaluation framework was also added using:
• Runtime
• Nodes Expanded
• Nodes Generated
• Maximum Frontier Size
• Path Cost

These metrics allow objective comparison between different search strategies.

4. Previous Status Compared with Current Progress

Previous Status:
• Basic architecture completed.
• Initial puzzle modelling completed.
• Search algorithms under development.
• Visualization and evaluation modules incomplete.

Current Status:
• Multiple puzzle environments integrated.
• Search algorithms fully implemented.
• Visualization framework completed.
• Performance analysis added.
• Documentation and presentation materials prepared.

5. Challenges Addressed

During this development period, several technical challenges were solved.

Algorithm Integration:
A common framework was created so different search algorithms could work with multiple puzzle environments.

State Representation:
A flexible modelling approach was developed to maintain compatibility between different puzzle types.

Result Interpretation:
Visualization and performance metrics were introduced to make algorithm behaviour easier to analyse.

6. Remaining Work and Future Plan

Before final submission, the remaining activities include final testing, improving visual presentation, completing documentation, and preparing the final demonstration.

Future improvements may include additional puzzle environments, advanced optimization techniques, improved animations, and online deployment.

7. Conclusion

The project has achieved significant progress after the previous update. The system has evolved into a complete AI-based puzzle-solving framework integrating multiple search algorithms, visualization, and performance evaluation.

The current implementation is ready for final demonstration and evaluation.